

Date: _____

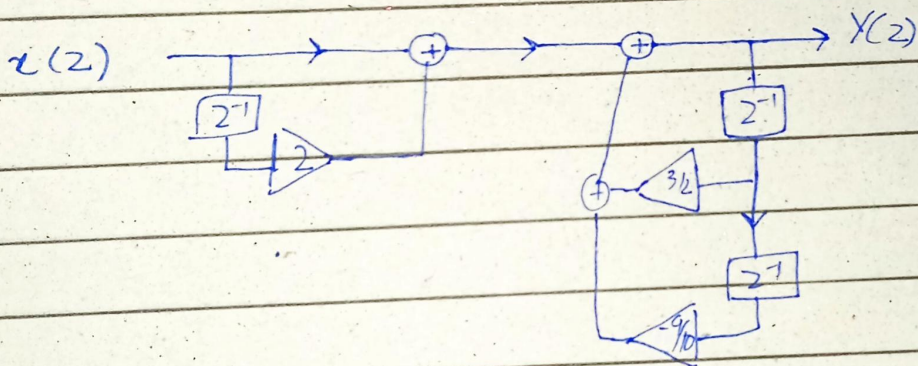
Muhammad Ahmad

FA23-BCE-113

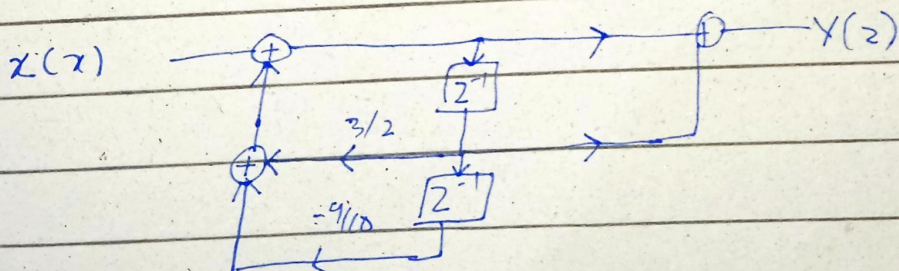
Q1: $H(z) = \frac{1 + 2z^{-1}}{1 - 1.5z^{-1} + 0.9z^{-2}}$

$$\frac{1 + 2z^{-1}}{1 - \frac{3}{2}z^{-1} + \frac{9}{10}z^{-2}}$$

Direct form - 1:



Direct form - 2:

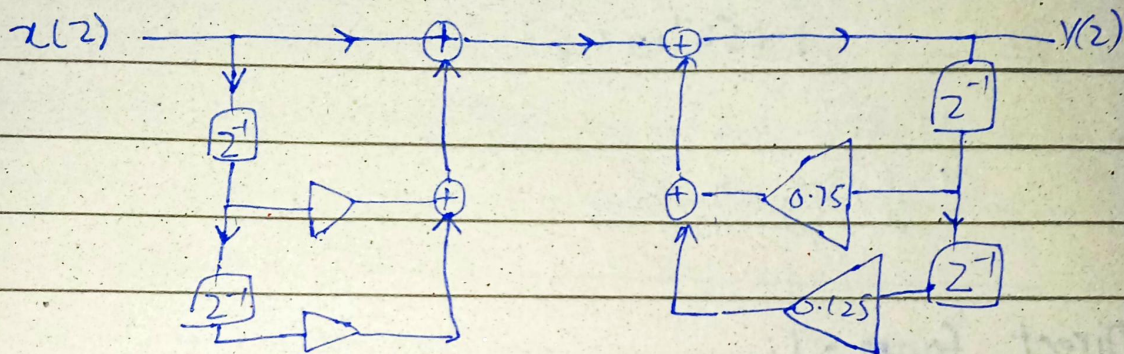


Date: _____

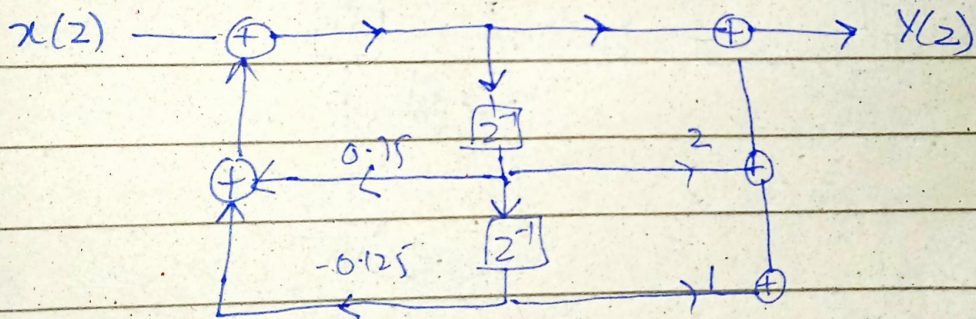
Q 2:

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

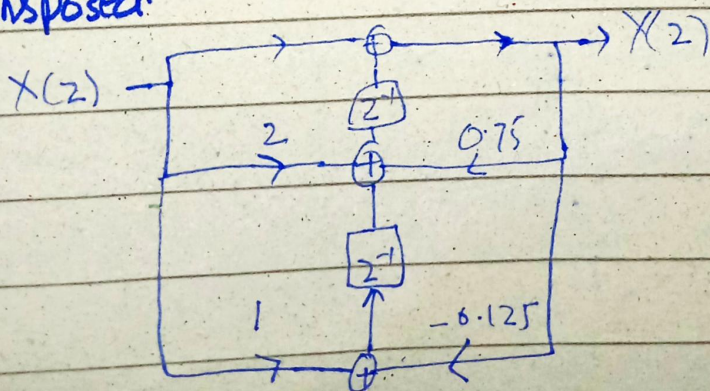
Direct form 1:



Direct form 2:



Transposed:



Date: _____

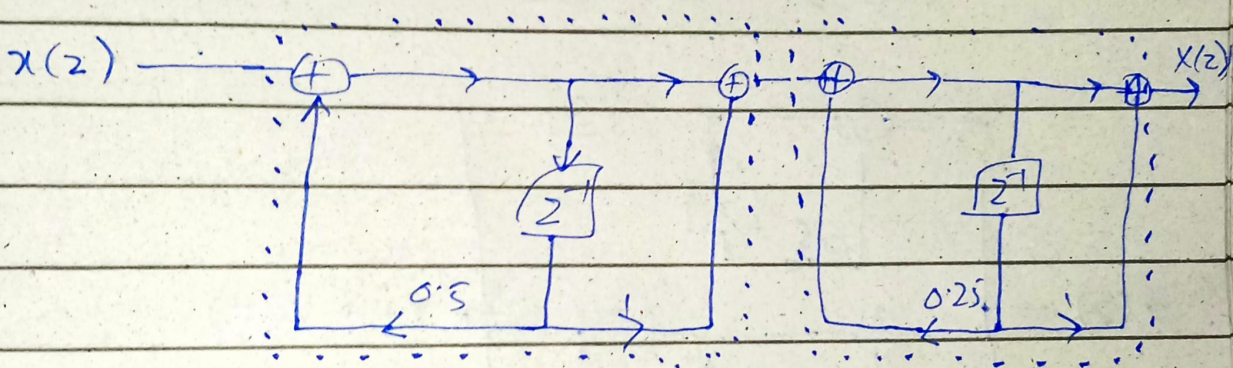
Cascaded

$$H(z) = \frac{1 + z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

$$= \frac{1 + z^{-1} + z^{-2}}{1 - 0.5z^{-1} - 0.25z^{-1} + 0.125z^{-2}}$$

$$\neq 0.5z^{-1} - 0.25z^{-1} + 0.125z^{-2}$$

$$= \left(\frac{1 + z^{-1}}{1 - 0.5z^{-1}} \right) \left(\frac{1 + z^{-1}}{1 - 0.25z^{-1}} \right)$$



Parallel:

$$H(z) = \frac{1 + z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}$$

$$8 + \frac{8z^{-1} - 7}{0.125z^{-2} - 0.75z^{-1} + 1}$$

$$0.125z^{-2} - 0.75z^{-1} + 1$$

$$8 + \frac{8z^{-1} - 7}{(0.5z^{-1} - 1)(0.25z^{-1} - 1)}$$

$$(0.5z^{-1} - 1)(0.25z^{-1} - 1)$$

①

Date: _____

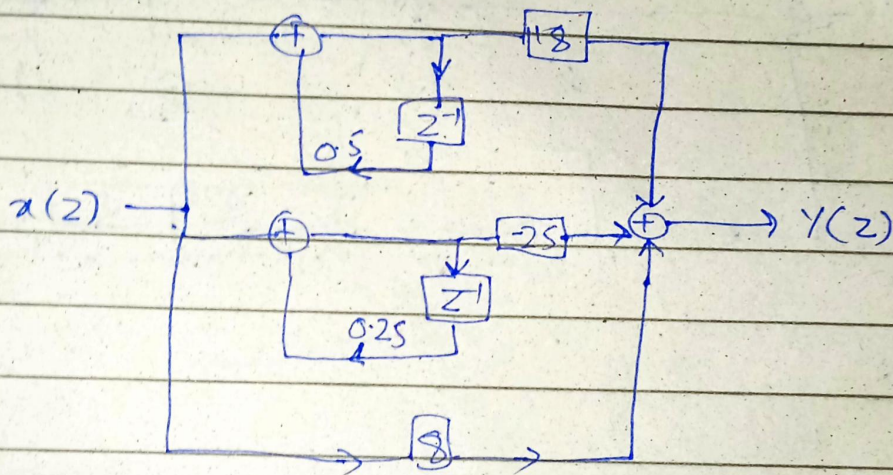
$$8z^{-1} - 7 = A(1 - 0.25z^{-1}) + B(1 - 0.5z^{-1})$$

$$A = 18$$

$$B = -25$$

now equation 2 become:

$$8 + \frac{18}{1 - 0.5z^{-1}} - \frac{25}{1 - 0.25z^{-1}}$$



Q.3:

(a) difference eq of given graph:

Direct form:

$$y[n] = 4y[n-1] + 7y[n-3] + 2y[n-4] - x[n]$$

Date: _____

Cascade form:

$$w[n] = x[n] + 3w[n-1] + w[n-2]$$

$$y[n] = w[n] + y[n-1] + 2y[n-2]$$

(b) System junction:

→ Difference eq:

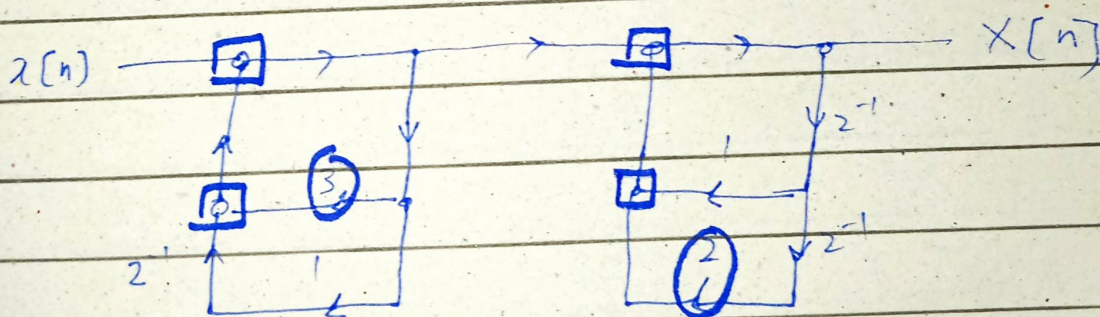
$$y[n] - 4y[n-1] + 7y[n-3] + 2y[n-4] = x[n]$$

→ Z transform:

$$Y(z) (1 - 4z^{-1} + 7z^{-3} + 2z^{-4}) = X(z)$$

$$H(z) = \frac{1}{1 - 4z^{-1} + 7z^{-3} + 2z^{-4}}$$

(c) Multiples and adder count:



Real Multiplications: 2

Real Additions: 4

Date: _____

(d) No, as it requires at least four delays to implement a fourth-order system so it is not possible to reduce number of storage registers.